CICSA PROJECT

WAZUH SIEM IMPLEMENTATION FOR SECURITY MONITORING

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Abstract

This project demonstrates the installation, configuration, and implementation of a Wazuh SIEM lab environment for centralized log collection and security monitoring. The Wazuh Manager is deployed on an Ubuntu system, while Linux and Windows machines are configured as Wazuh agents. System, authentication, and security logs are collected and analyzed in real time through the Wazuh Dashboard. The project helps in understanding SIEM concepts, log monitoring, alert generation, and basic threat detection techniques.

Introduction

Wazuh is an open-source Security Information and Event Management (SIEM) and Extended Detection and Response (XDR) platform. It is used to collect, analyze, and correlate logs from multiple systems to detect security threats, intrusions, and abnormal activities. This project focuses on building a small-scale Wazuh lab to simulate real-world security monitoring.

Objectives

- To install and configure Wazuh Manager on Ubuntu.
- To configure Linux and Windows systems as Wazuh agents.
- To collect and analyze system and security logs.
- To detect authentication failures and security events.
- To understand SIEM workflow and alerting mechanisms.

Tools and Technologies

- Wazuh SIEM – Used for centralized log monitoring, security event analysis, and threat detection.
- Ubuntu Server – Used to install and run the Wazuh Manager.
- Kali Linux – Used as the attacker machine to simulate security attacks.
- Hydra – Used to perform brute-force attack simulation.
- VirtualBox – Used to create and manage virtual machines for the lab environment.
- Windows OS – Used as an endpoint system monitored by the Wazuh agent.

Lab Architecture

- **Wazuh Manager:** Installed on Ubuntu server.
- **Linux Agent:** Kali Linux system.
- **Windows Agent:** Windows 10 system.
- **Wazuh Dashboard:** Web-based interface for monitoring.
- **Description:** Agents forward logs securely to the Manager, which uses predefined rules to display alerts.

System Requirements

- **Hardware:** 8GB RAM, 2 Cores CPU, 50GB Disk.
- **Software:** Ubuntu 22.04, Kali Linux, Windows 10, VirtualBox.

Installation & Configuration

- **Manager:** Installed using few commands.
- **Agents:** Installed and linked to Manager IP via configuration files.

Log Collection

Wazuh agent was configured to continuously monitor system and authentication logs and securely forward the collected events to the Wazuh Manager for centralized analysis and correlation.

- Linux: Monitoring /var/log/auth.log and /var/log/syslog for authentication attempts and system activities.
- Windows: Monitoring Security Event Logs, specifically login and authentication-related events.

Testing & Results

- **Test:** Failed SSH logins on Linux & Invalid logins on Windows.
- **Result:** Alerts appeared in real-time on the Dashboard with severity levels.

Methodology

The implementation of the Wazuh SIEM monitoring environment was carried out through a series of systematic steps. Initially, the Wazuh Manager was installed and configured on an Ubuntu server to act as the central log analysis system. Subsequently, Wazuh Agents were installed on the monitored endpoints, including Linux and Windows systems, to collect security events and system logs. After successful agent registration, the agents were configured to forward log data securely to the Wazuh Manager. Finally, attack simulations such as SSH brute-force attempts were performed to generate security events, which were analyzed through the Wazuh dashboard for threat detection and monitoring.

INSTALLATION OF MONITORING TOOL: WAZUH SIEM

The installation process is divided into two main phases: the deployment of the central **Wazuh Manager** on a Linux server and the configuration of **Wazuh Agents** on the endpoints (Linux and Windows).

- **WAZUH MANAGER INSTALLATION (Ubuntu)**

The Wazuh Manager is the heart of the SIEM, responsible for analyzing data received from agents

- **WAZUH AGENT INSTALLATION** (Monitored Systems - Linux & Windows)

Wazuh Manager Installation (Ubuntu)

System Update and Preparation *Before starting the installation, ensure the system packages are up to date to avoid dependency conflicts***.**

- ***sudo apt-get update && sudo apt-get upgrade -y***

```
File  Edit  View  Search  Terminal  Help
oot@whoami:/home/sreekuttan# sudo apt-get update && sudo apt-get upgrade -y
gn:1 http://dl.google.com/linux/chrome-remote-desktop/deb stable InRelease
it:2 https://packages.wazuh.com/4.x/apt stable InRelease
it:3 https://packages.microsoft.com/ubuntu/22.04/mssql-server-2022 jammy InRelease
it:4 https://packages.microsoft.com/repos/code stable InRelease
it:5 http://dl.google.com/linux/chrome-remote-desktop/deb stable Release
it:6 http://in.archive.ubuntu.com/ubuntu jammy InRelease
it:7 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
it:8 http://security.ubuntu.com/ubuntu jammy-security InRelease
it:9 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
rr:3 https://packages.microsoft.com/ubuntu/22.04/mssql-server-2022 jammy InRelease
 The following signatures couldn't be verified because the public key is not available: NO_PUBKEY EB3E94ADBE1229CF
it:10 https://ppa.launchpadcontent.net/mozillateam/ppa/ubuntu jammy InRelease
et:12 https://pkgs.tailscale.com/stable/ubuntu jammy InRelease
etched 6,581 B in 2s (3,703 B/s)
eading package lists... Done
: An error occurred during the signature verification. The repository is not updated and the previous index files will be used. GPG error: https://packages.microsoft.c
/ubuntu/22.04/mssql-server-2022 jammy InRelease: The following signatures couldn't be verified because the public key is not available: NO_PUBKEY EB3E94ADBE1229CF
: Failed to fetch https://packages.microsoft.com/ubuntu/22.04/mssql-server-2022/dists/jammy/InRelease  The following signatures couldn't be verified because the public
ey is not available: NO_PUBKEY EB3E94ADBE1229CF
: Some index files failed to download. They have been ignored, or old ones used instead.
eading package lists... Done
uilding dependency tree... Done
eading state information... Done
alculating upgrade... Done
he following packages have been kept back:
 linux-generic-hwe-22.04 linux-headers-generic-hwe-22.04 linux-image-generic-hwe-22.04
he following packages will be upgraded:
 chrome-remote-desktop libc-bin libc-dev-bin libc-devtools libc6 libc6:i386 libc6-dbg libc6-dev libc6-i386 libfreerdp-client2-2 libfreerdp-server2-2 libfreerdp2-2
 libldap-2.5-0 libldap-common libmysqlclient21 libwinpr2-2 locales mysql-client-8.0 mysql-client-core-8.0 mysql-server mysql-server-8.0 mysql-server-core-8.0
 thunderbird thunderbird-gnome-support thunderbird-locale-en thunderbird-locale-en-us
6 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
eed to get 158 MB of archives.
fter this operation, 338 kB disk space will be freed.
et:1 http://dl.google.com/linux/chrome-remote-desktop/deb stable/main amd64 chrome-remote-desktop amd64 145.0.7632.9 [18.8 MB]
et:2 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libc6-i386 amd64 2.35-0ubuntu3.13 [2,837 kB]
et:3 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libc-devtools amd64 2.35-0ubuntu3.13 [29.0 kB]
```

Wazuh Installation command

- ***curl -sO https://packages.wazuh.com/4.11/wazuh-install.sh && curl -sO https://packages.wazuh.com/4.11/config.yml***
- ***head -n 5 wazuh-install.sh***

```
root@whoami:/home/sreekuttan# curl -sO https://packages.wazuh.com/4.11/wazuh-install.sh && curl -sO https://packages.wazuh.com/4.11/config.yml
root@whoami:/home/sreekuttan# head -n 5 wazuh-install.sh
#!/bin/bash

# Wazuh installer
# Copyright (C) 2015, Wazuh Inc.
#
```

- ***bash wazuh-install.sh -a -o***

```
root@whoami:/home/sreekuttan# bash wazuh-install.sh -a -o
4/02/2026 12:46:30 INFO: Starting Wazuh installation assistant. Wazuh version: 4.11.2 (x86_64/AMD64)
4/02/2026 12:46:30 INFO: Verbose logging redirected to /var/log/wazuh-install.log
4/02/2026 12:46:40 INFO: --- Removing existing Wazuh installation ---
4/02/2026 12:46:40 INFO: Installation cleaned.
4/02/2026 12:46:40 INFO: Verifying that your system meets the recommended minimum hardware requirements.
4/02/2026 12:46:41 INFO: Wazuh web interface port will be 443.
4/02/2026 12:46:58 INFO: Wazuh repository added.
4/02/2026 12:46:58 INFO: --- Configuration files ---
4/02/2026 12:46:58 INFO: Generating configuration files.
4/02/2026 12:46:59 INFO: Generating the root certificate.
4/02/2026 12:46:59 INFO: Generating Admin certificates.
4/02/2026 12:47:00 INFO: Generating Wazuh indexer certificates.
4/02/2026 12:47:00 INFO: Generating Filebeat certificates.
4/02/2026 12:47:00 INFO: Generating Wazuh dashboard certificates.
4/02/2026 12:47:01 INFO: Created wazuh-install-files.tar. It contains the Wazuh cluster key, certificates, and passwords necessary for installation.
4/02/2026 12:47:01 INFO: --- Wazuh indexer ---
4/02/2026 12:47:01 INFO: Starting Wazuh indexer installation.
4/02/2026 12:52:26 INFO: Wazuh indexer installation finished.
4/02/2026 12:52:27 INFO: Wazuh indexer post-install configuration finished.
4/02/2026 12:52:27 INFO: Starting service wazuh-indexer.
4/02/2026 12:53:24 INFO: wazuh-indexer service started.
4/02/2026 12:53:24 INFO: Initializing Wazuh indexer cluster security settings.
4/02/2026 12:53:50 INFO: Wazuh indexer cluster security configuration initialized.
4/02/2026 12:53:50 INFO: Wazuh indexer cluster initialized.
4/02/2026 12:53:50 INFO: --- Wazuh server ---
4/02/2026 12:53:50 INFO: Starting the Wazuh manager installation.
4/02/2026 12:57:25 INFO: Wazuh manager installation finished.
4/02/2026 12:57:26 INFO: Wazuh manager vulnerability detection configuration finished.
4/02/2026 12:57:26 INFO: Starting service wazuh-manager.
4/02/2026 12:57:59 INFO: wazuh-manager service started.
4/02/2026 12:58:00 INFO: Starting Filebeat installation.
4/02/2026 13:04:43 INFO: Filebeat installation finished.
4/02/2026 13:04:51 INFO: Filebeat post-install configuration finished.
4/02/2026 13:04:51 INFO: Starting service filebeat.
```

Saving Credentials *Once the installation is complete, the terminal will display the* ***Admin Username*** *and* ***Password****. It is critical to copy and save these credentials, as they are required to access the web dashboard.*

Verifying Services To ensure the manager is running correctly, check the status of the Wazuh manager service

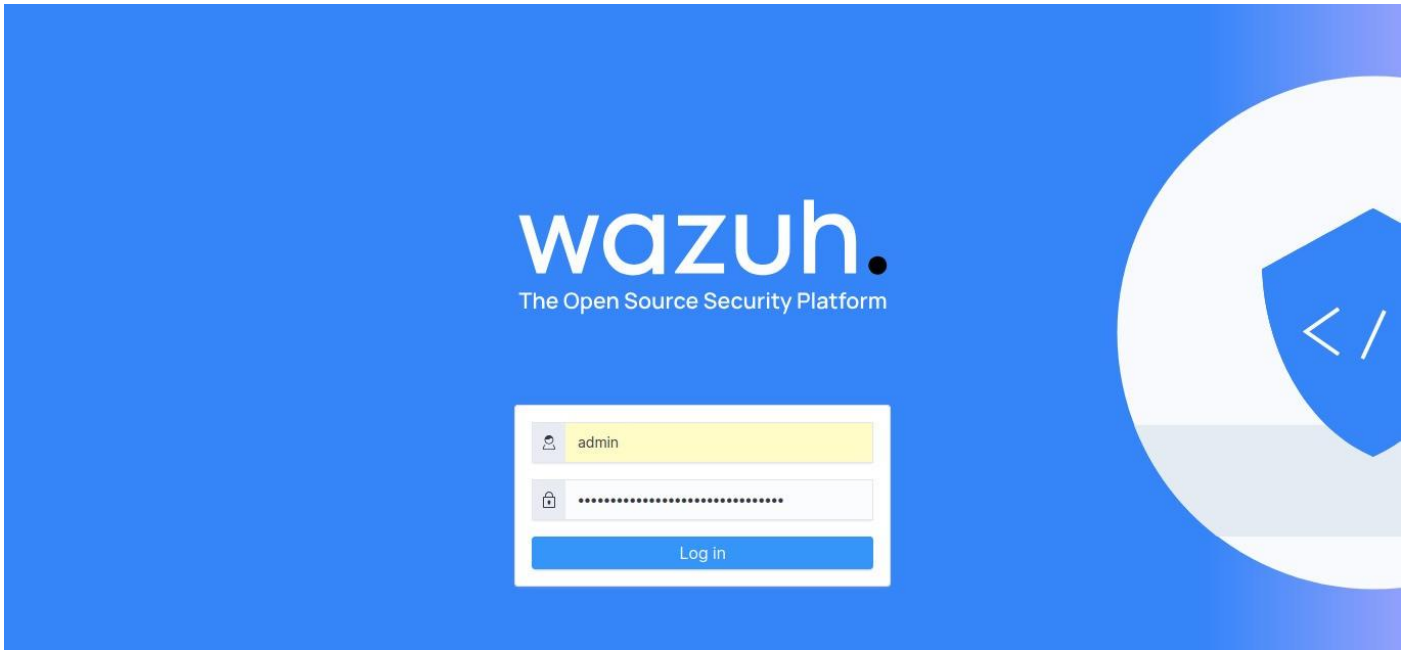
- ***sudo systemctl status wazuh-manager***

```
root@whoami: ~                    root@whoami: /home/sreekuttan
root@whoami:/home/sreekuttan# sudo systemctl status wazuh-manager
● wazuh-manager.service - Wazuh manager
     Loaded: loaded (/lib/systemd/system/wazuh-manager.service; enabled; vendor preset: enabled)
     Active: active (running) since Wed 2026-02-04 16:19:10 +0545; 23min ago
      Tasks: 141 (limit: 6910)
     Memory: 346.3M
        CPU: 2min 13.836s
     CGroup: /system.slice/wazuh-manager.service
             ├─59729 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh-apid.py
             ├─59768 /var/ossec/bin/wazuh-authd
             ├─59784 /var/ossec/bin/wazuh-db
             ├─59796 /var/ossec/bin/wazuh-execd
             ├─59810 /var/ossec/bin/wazuh-analysisd
             ├─59823 /var/ossec/bin/wazuh-syscheckd
             ├─59840 /var/ossec/bin/wazuh-remoted
             ├─59851 /var/ossec/bin/wazuh-logcollector
             ├─59864 /var/ossec/bin/wazuh-monitord
             ├─59876 /var/ossec/bin/wazuh-modulesd
             ├─60443 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh-apid.py
             ├─60446 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh-apid.py
             └─60449 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh-apid.py

Feb 04 16:19:00 whoami env[59669]: Started wazuh-db...
Feb 04 16:19:01 whoami env[59669]: Started wazuh-execd...
Feb 04 16:19:02 whoami env[59669]: Started wazuh-analysisd...
Feb 04 16:19:03 whoami env[59669]: Started wazuh-syscheckd...
Feb 04 16:19:05 whoami env[59669]: Started wazuh-remoted...
Feb 04 16:19:06 whoami env[59669]: Started wazuh-logcollector...
Feb 04 16:19:07 whoami env[59669]: Started wazuh-monitord...
Feb 04 16:19:08 whoami env[59669]: Started wazuh-modulesd...
Feb 04 16:19:10 whoami env[59669]: Completed.
Feb 04 16:19:10 whoami systemd[1]: Started Wazuh manager.
root@whoami:/home/sreekuttan#
```

Accessing The Wazuh Dashboard

Web Interface Login Open a web browser and enter the IP address of your manager using HTTPS:

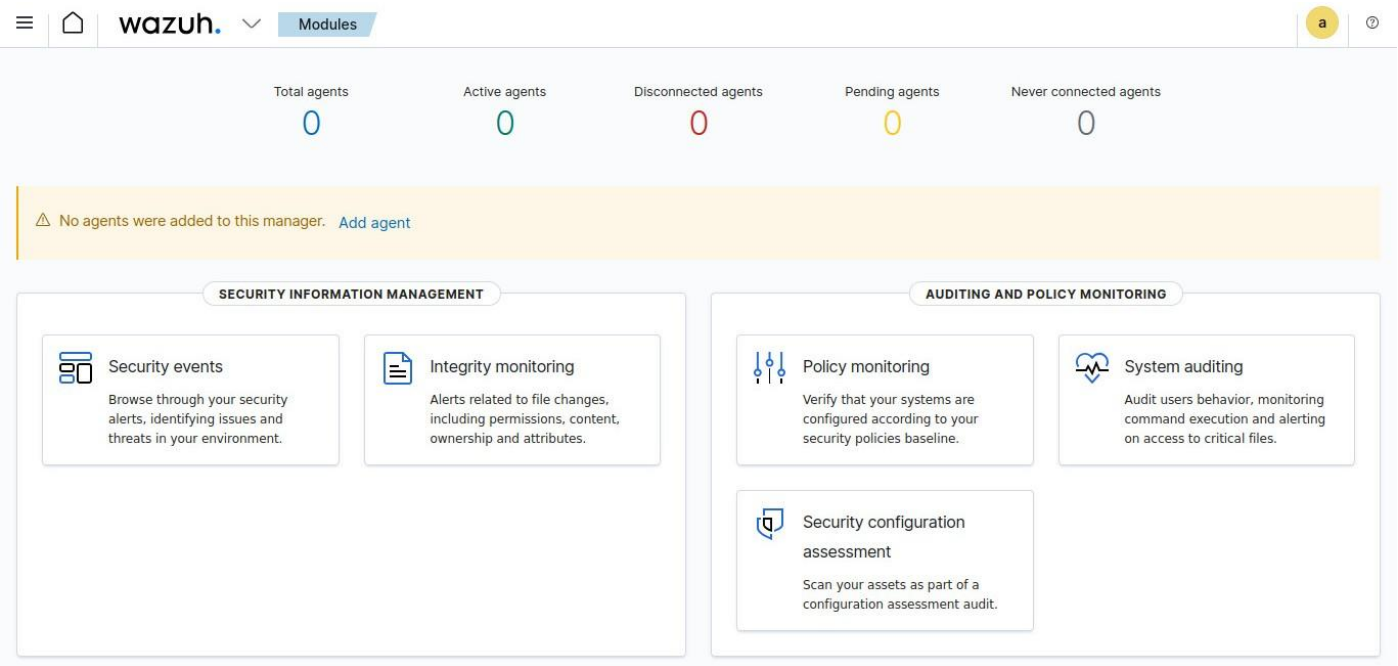
- **URL: https://ip**
- **Username: admin**
- **Password: XYZ**



Dashboard Overview

After a successful login, the Wazuh Dashboard home page appears. From here, you can navigate to the

Agents tab to deploy and monitor new endpoints.



Installation Of Monitoring Tool: Wazuh Agent

To enable security monitoring, the Wazuh agent must be installed on the endpoints. These agents are responsible for scanning the system for vulnerabilities, monitoring log files, and sending the data to the central Wazuh Manager.

Wazuh Agent Installation on Kali

Step 1: Download and Install the Agent The agent is installed by downloading the package and assigning the Manager's IP address during the installation process.

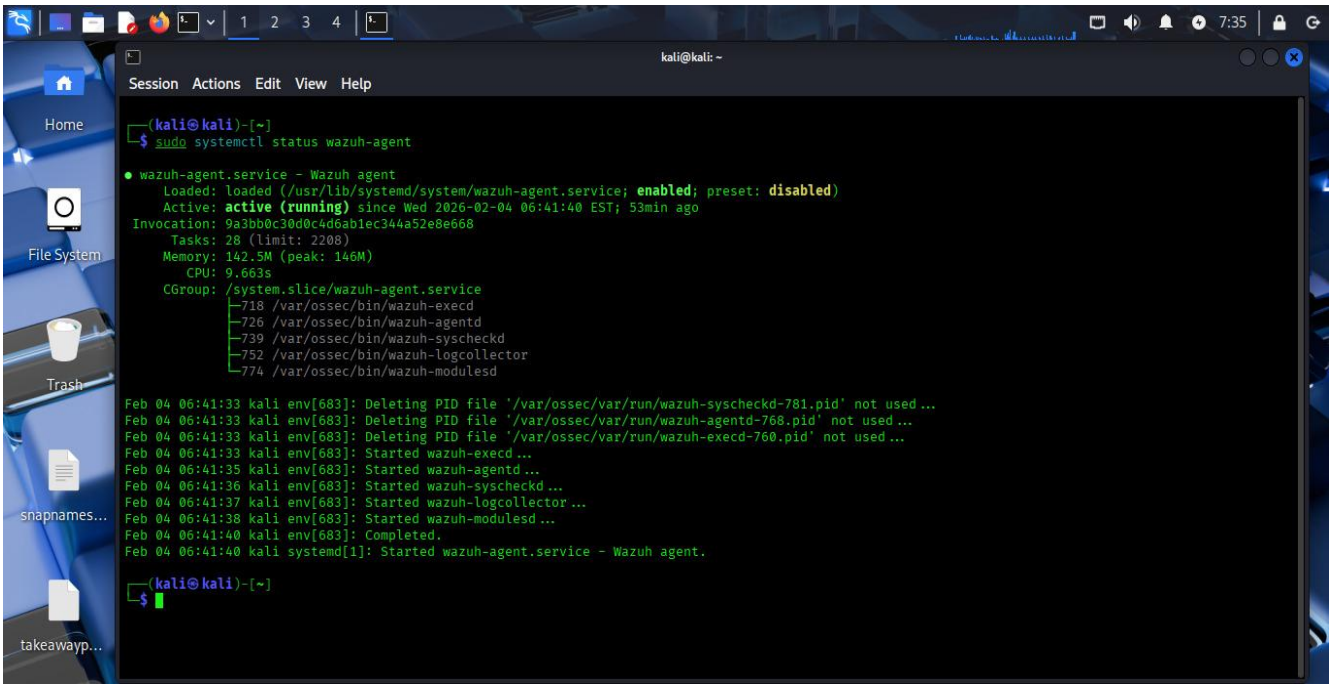
- ***wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.x.x-1_amd64.deb***
- ***sudo WAZUH_MANAGER='<MANAGER_IP>' dpkg -i wazuh-agent_4.x.x-1_amd64.deb***

Step 2: Start and Enable the Agent Service After installation, the agent service must be started and enabled to ensure it runs automatically on system boot.

- ***sudo systemctl daemon-reload***
- ***sudo systemctl enable wazuh-agent***
- ***sudo systemctl start wazuh-agent***

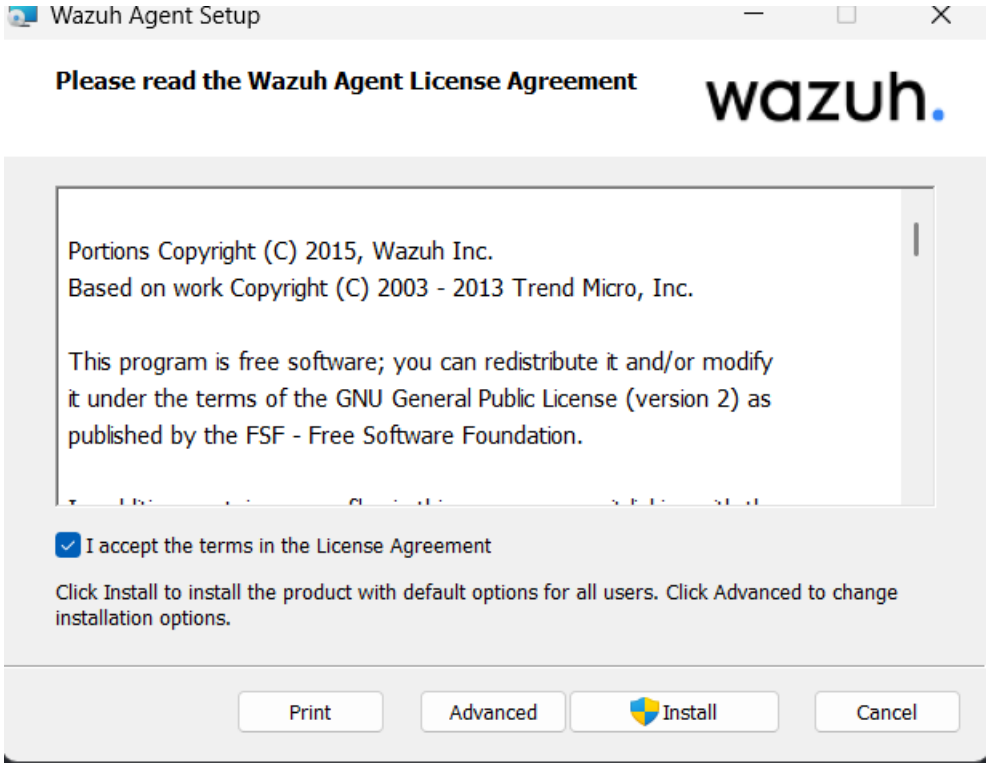
Step 3: Verify Connection Check the status to ensure the agent is successfully communicating with the manager

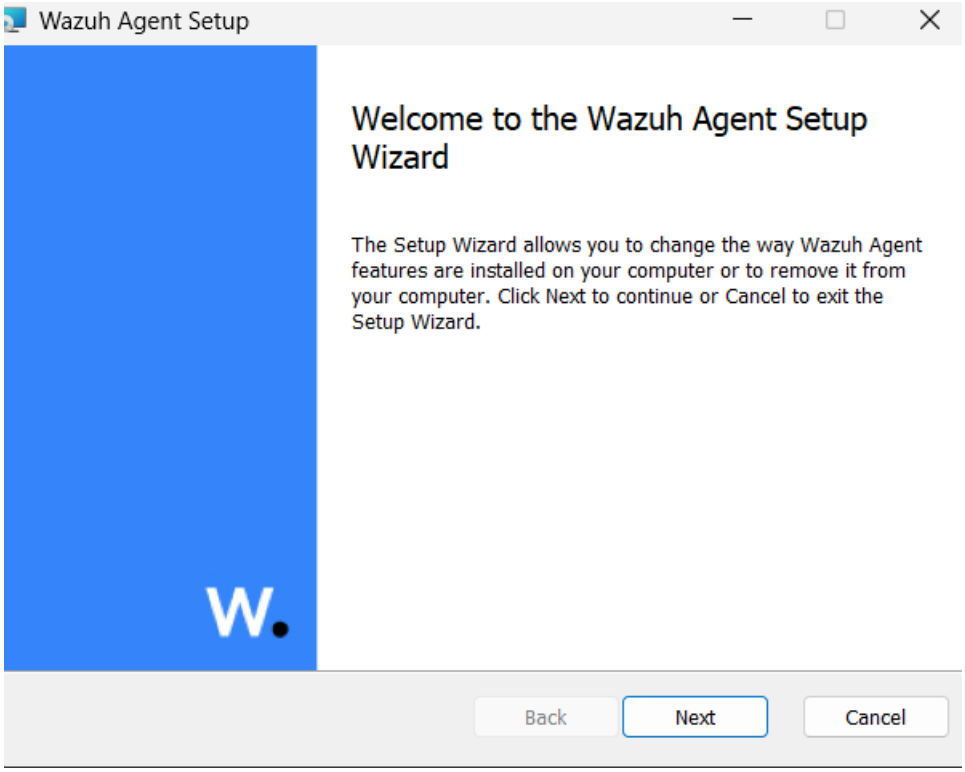
- ***sudo systemctl status wazuh-agent***



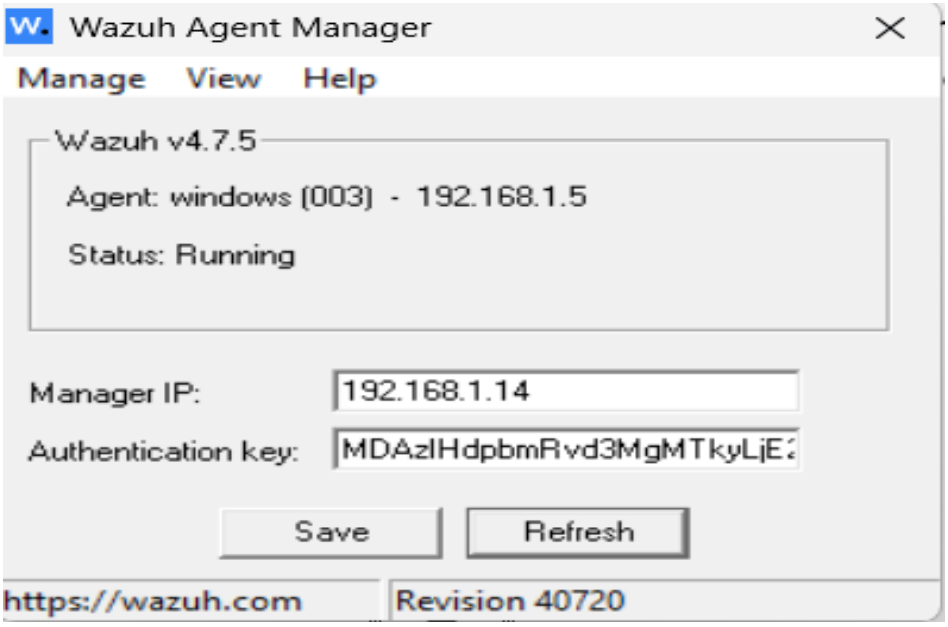
Wazuh Agent Installation on windows

- **Download**: Wazuh agent MSI installer.
- **Run Installer**: Double-click the .msi file and : Click "INSTALL" through the prompts





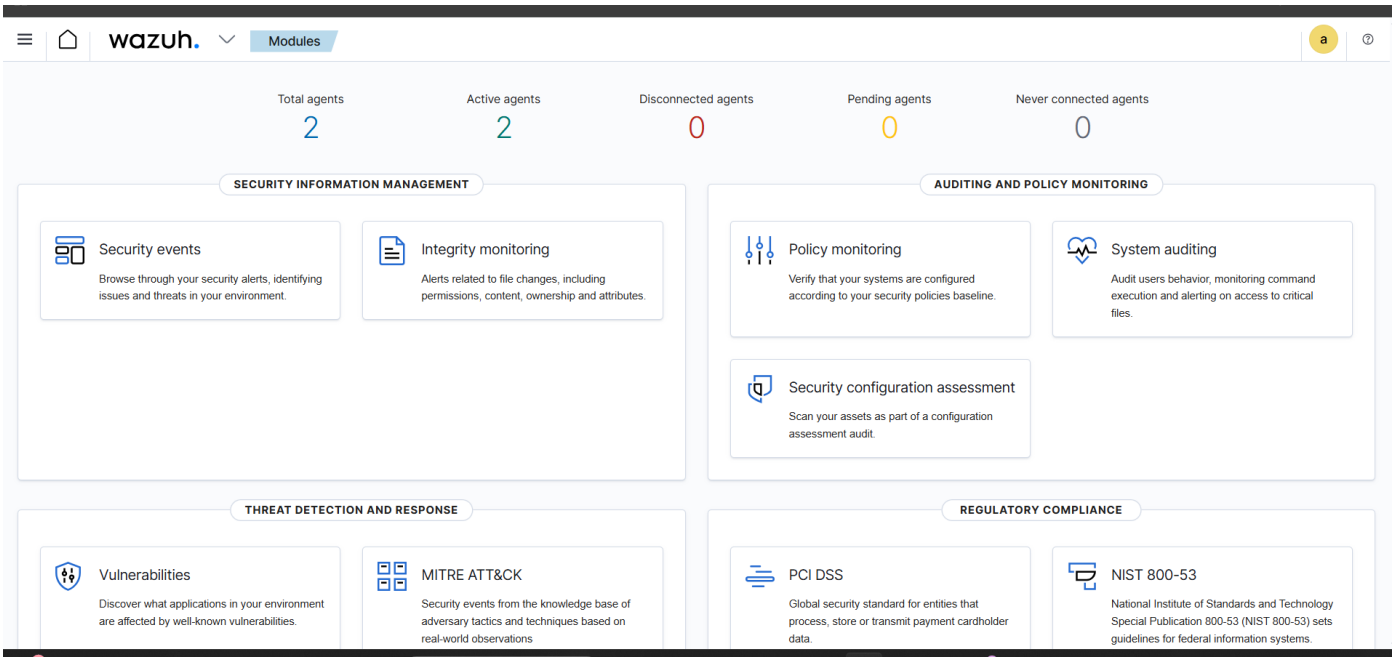
- Enter the **Manager IP**
- Enter the **Authentication Key**
- **Save and Start**: Click **Manage** > **Start**



After wazuh agents successfully install

Once the agents are installed on Linux and windows , they must be verified on the central dashboard.

Access the Wazuh Dashboard via web browser.

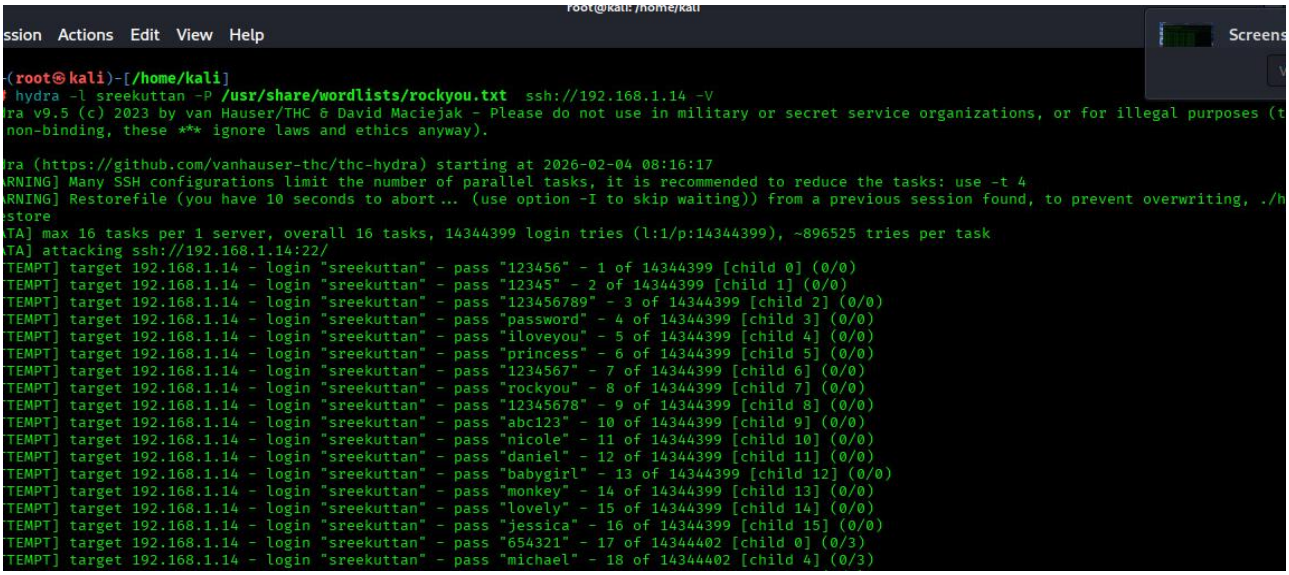


Attack Simulation (Brute Force)

Target: Kali Linux (Agent)

Step 1: Execute the Attack From your attacker machine, try to connect to the Kali Linux agent via SSH using random, incorrect passwords multiple times.

Command used: hydra -l username -P /usr/share/wordlists/rockyou.txt ssh:// IP -V



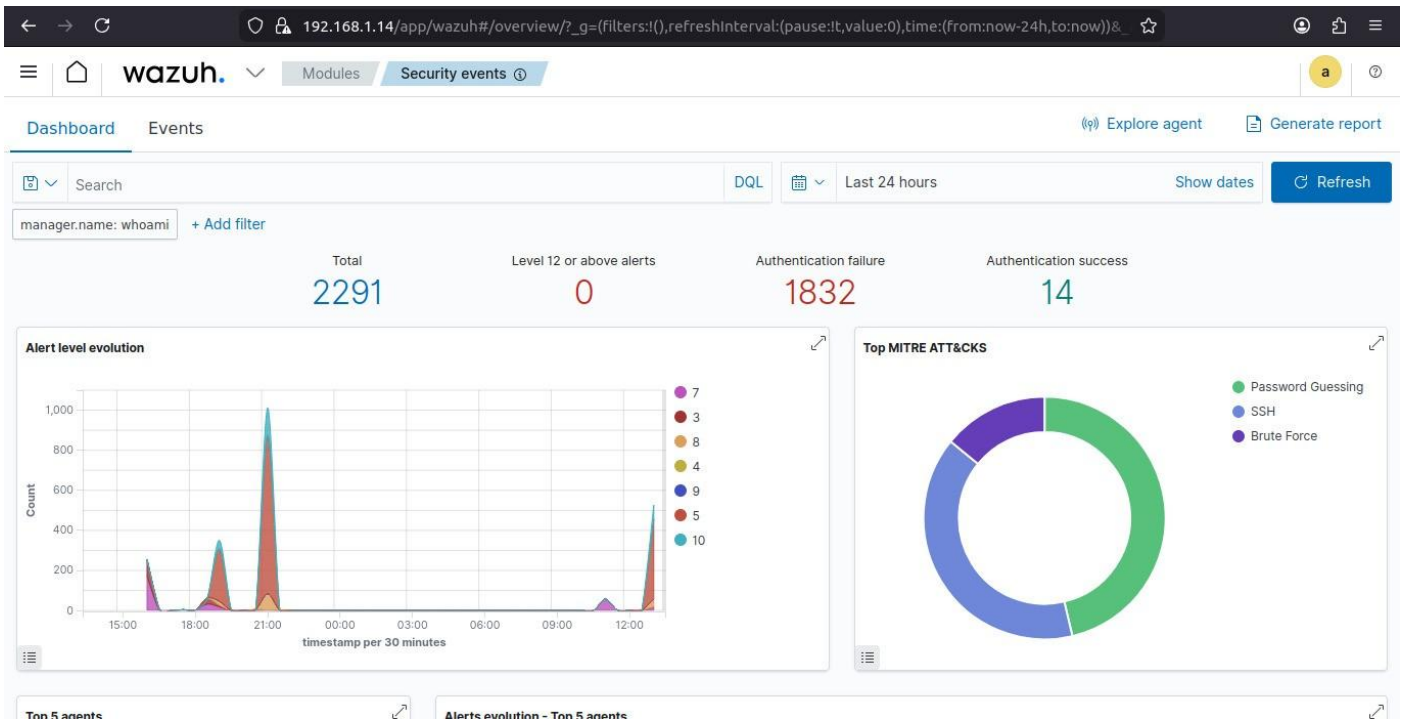
```
ssion  Actions  Edit  View  Help
┌──(root㉿kali)-[/home/kali]
└─# hydra -l sreekuttan -P /usr/share/wordlists/rockyou.txt  ssh://192.168.1.14 -V
ra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (t
non-binding, these ** ignore laws and ethics anyway).

ra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-02-04 08:16:17
RNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
RNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a previous session found, to prevent overwriting, ./h
store
TA] max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries (l:1/p:14344399), ~896525 tries per task
TA] attacking ssh://192.168.1.14:22/
TEMPT] target 192.168.1.14 - login "sreekuttan" - pass "123456" - 1 of 14344399 [child 0] (0/0)
TEMPT] target 192.168.1.14 - login "sreekuttan" - pass "12345" - 2 of 14344399 [child 1] (0/0)
TEMPT] target 192.168.1.14 - login "sreekuttan" - pass "123456789" - 3 of 14344399 [child 2] (0/0)
TEMPT] target 192.168.1.14 - login "sreekuttan" - pass "password" - 4 of 14344399 [child 3] (0/0)
TEMPT] target 192.168.1.14 - login "sreekuttan" - pass "iloveyou" - 5 of 14344399 [child 4] (0/0)
TEMPT] target 192.168.1.14 - login "sreekuttan" - pass "princess" - 6 of 14344399 [child 5] (0/0)
TEMPT] target 192.168.1.14 - login "sreekuttan" - pass "1234567" - 7 of 14344399 [child 6] (0/0)
TEMPT] target 192.168.1.14 - login "sreekuttan" - pass "rockyou" - 8 of 14344399 [child 7] (0/0)
TEMPT] target 192.168.1.14 - login "sreekuttan" - pass "12345678" - 9 of 14344399 [child 8] (0/0)
TEMPT] target 192.168.1.14 - login "sreekuttan" - pass "abc123" - 10 of 14344399 [child 9] (0/0)
TEMPT] target 192.168.1.14 - login "sreekuttan" - pass "nicole" - 11 of 14344399 [child 10] (0/0)
TEMPT] target 192.168.1.14 - login "sreekuttan" - pass "daniel" - 12 of 14344399 [child 11] (0/0)
TEMPT] target 192.168.1.14 - login "sreekuttan" - pass "babygirl" - 13 of 14344399 [child 12] (0/0)
TEMPT] target 192.168.1.14 - login "sreekuttan" - pass "monkey" - 14 of 14344399 [child 13] (0/0)
TEMPT] target 192.168.1.14 - login "sreekuttan" - pass "lovely" - 15 of 14344399 [child 14] (0/0)
TEMPT] target 192.168.1.14 - login "sreekuttan" - pass "jessica" - 16 of 14344399 [child 15] (0/0)
TEMPT] target 192.168.1.14 - login "sreekuttan" - pass "654321" - 17 of 14344402 [child 0] (0/3)
TEMPT] target 192.168.1.14 - login "sreekuttan" - pass "michael" - 18 of 14344402 [child 4] (0/3)
```

Monitoring and Threat Detection

- **Alert Generation:** The Wazuh Manager successfully captured the failed login attempts.

- **Security Events:** Navigated to the **Security Events** module on the Wazuh Dashboard.

  The Wazuh Manager analyzed collected logs using built-in decoders and rules. Based on repeated failed SSH login attempts, the system triggered high-severity alerts indicating a brute force attack.



Wazuh Security Events dashboard showing detection of SSH brute force attack

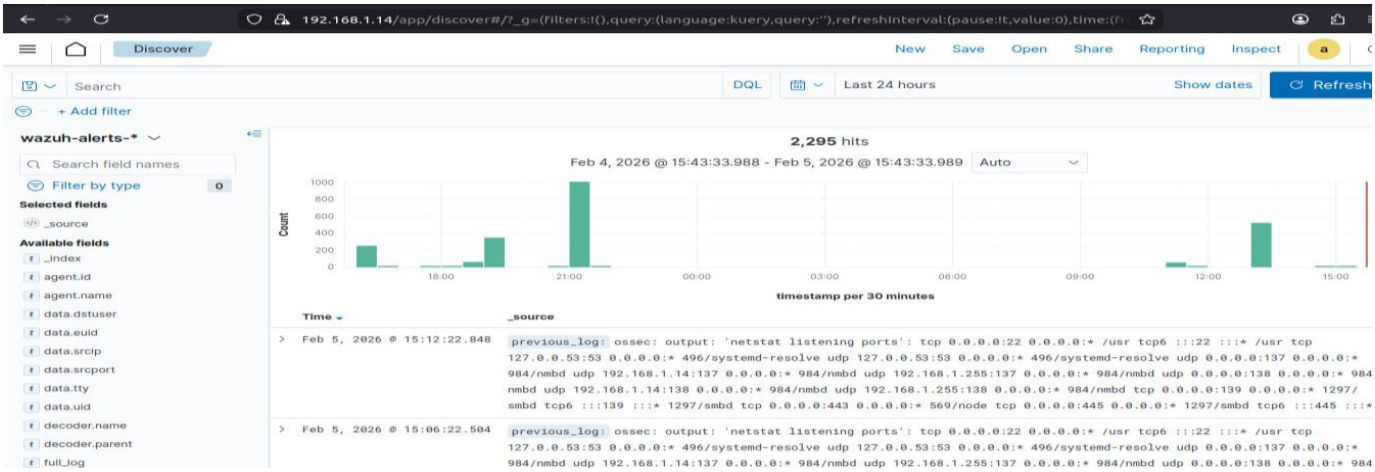
Findings:

- **Total Alerts:** 2291 alerts were triggered during the attack.
- **Authentication Failures:** A total of 1832 failed login attempts
- **Alert Level Evolution:** The dashboard graph showed a significant spike at the time of the attack (around 21:00), indicating a high-intensity brute force attempt.

- **MITRE ATTACK Mapping:** Wazuh automatically mapped the detected activity to the **MITRE ATTACK technique T1110 (Brute Force)**, classifying it as an authentication-based attack.

Threat Analysis

The detected alerts were analyzed using the Wazuh dashboard. A high number of failed SSH login attempts were observed from a single source IP, indicating a brute force attack. The alert severity levels increased significantly during the attack period. A small number of successful login attempts were also observed, highlighting the potential risk of credential compromise. The activity was mapped to the MITRE ATTACK technique T1110 (Brute Force).



Remediation

- *Implement account lockout policies for repeated failed logins.*
- *Enable two-factor authentication (2FA) for critical accounts.*
- *Keep systems updated with the latest security patches.*
- *Use strong password policies to prevent weak credentials.*
- *Restrict SSH access to trusted IP addresses.*

Challenges Faced

- *Network Connectivity: Ensuring the Windows and Kali VMs could communicate with the Ubuntu Manager (often required switching VirtualBox settings to 'Bridged Adapter' or 'Host-Only').*
- *Resource Constraints: Running Ubuntu and Kali,and windows simultaneously requires significant RAM (minimum 8GB–12GB), which caused system lag during the simulation.*
- ***Firewall Blocking:** The Ubuntu Manager's firewall (UFW) initially blocked ports **1514** and **1515**, preventing the Agents from connecting until the ports were manually opened.*
- **Credential Management:** The auto-generated password for the 'admin' user is complex and had to be carefully retrieved from the installation logs.

Conclusion

*The project successfully demonstrated the implementation of **Wazuh SIEM** for centralized log management across a multi-platform environment . By simulating a Brute Force attack, we proved that Wazuh effectively detects and alerts on security threats in real-time. This setup enhances an organization's security posture by providing deep visibility into endpoint activities and ensuring faster incident response.*

References

- Wazuh Documentation – https://documentation.wazuh.com
- Wazuh Installation Guide – https://documentation.wazuh.com/current/installation-guide/
- Ubuntu Server Documentation – https://ubuntu.com/server/docs
- Hydra Tool Documentation – https://github.com/vanhauser-thc/thc-hydra
- MITRE ATT&CK Framework – https://attack.mitre.org